

HIKET — the storyline, v3

second pass: the arc fitted to the structure of the paper

Version of September 3, 2026. Numbers keyed to RUN_IDs 20260831_1624* (arm B). Built from the 19 annotations on v2 (revisions/HIKET_storyline_v2_annotated_20260903.pdf).

How to read this. The outer shell is now the conventional one — Introduction, Materials and methods, Results and Discussion, Conclusions — because that is what the paper has to be. **OCAR has not been abandoned; it has been moved inside.** The Opening and Challenge live in the Introduction, the Action is the Methods, and the Resolution is distributed across the Results in the order of importance set below. Each block is still one claim, its evidence, and nothing else. ▶ **name** marks a figure; ★ **TO BUILD: name** one still to build; □ **LIT REVIEW: topic** a literature review that must happen before the section can be written. Amber boxes are things that could sink a claim; green boxes are tests we could run; **violet boxes carry Lorenzo’s v2 annotations** and what was done about them.

national trend — the main story ~55%	ridge position ~25%	local variance ~20%
does a transient start reproduce the observed national trajectory, and what does fitting it require of the models?	does it matter where on the input/turnover ridge we land? (extrapolation)	what the models cannot see at all — raised as questions, not solved

Annotation (p17): “*your draft is a lot on the model perspective but it will have to be more anchored in biology in the end ... weight is about 50–60% general trends, 20–30% this side arc ... remaining unexplained local variance and what we could do about it (very briefly the last).*” **Done:** the budget above governs this version, and the biological content that the input result demands is now \$3.0.8 rather than four lines in a mechanism list.

Contents

1	Introduction	2
1.1	A sink that appears to be levelling off	2
1.2	Why it matters for the national greenhouse gas balance	3
1.3	The convention: soils initialised at steady state	3
1.4	Finnish forests are a managed landscape, not an equilibrium one	3
1.5	Objectives	4
2	Materials and methods	4
2.1	Data	4
2.2	The model ladder	6
2.3	Transient initialisation	6
2.4	A fair frame	6
2.5	Calibration and error model	6
2.6	Analyses	7

3 Results and discussion	7
Strand 1 — the national trajectory (55%)	7
Strand 2 — does ridge position matter? (25%)	13
Strand 3 — what none of the models can see (20%)	15
4 Conclusions	16
Open decisions and what is still missing	16

1 Introduction

1.1 A sink that appears to be levelling off

1.1.0.1 What the inventory measures. On the balanced plot set the mean stock runs $64.5 \rightarrow 71.5 \rightarrow 73.6 \text{ tC ha}^{-1}$ across the three campaigns: $+7.0$, then $+2.1$. Accumulation, bending over. ► [F3_mean_soc](#)

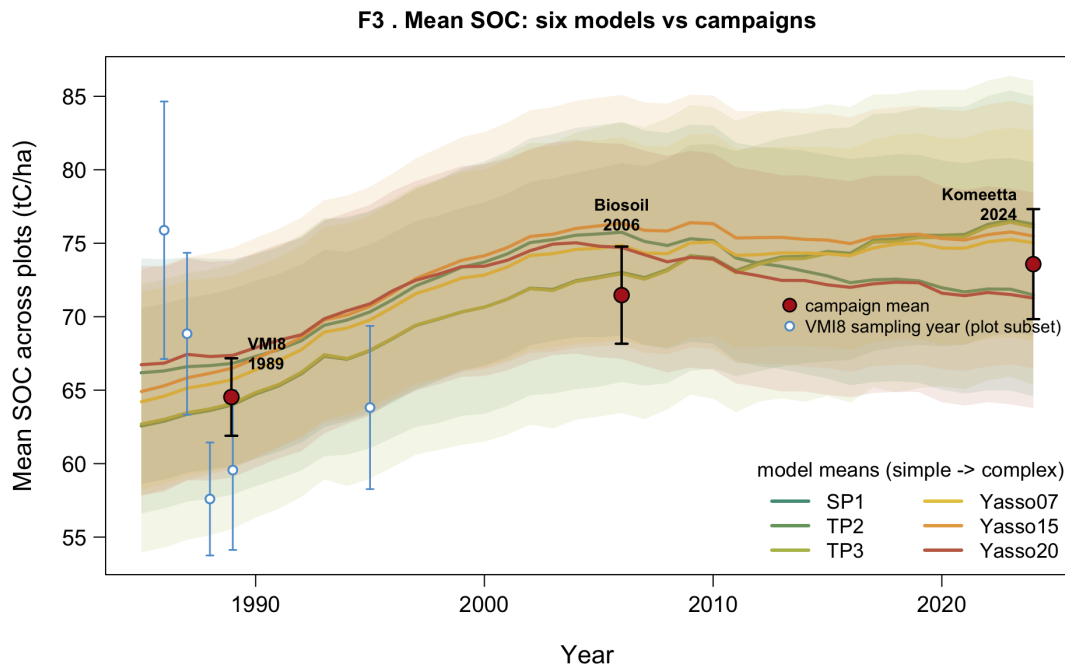


Figure 1: [F3_mean_soc](#)

1.1.0.2 The rate, and what it is a rate of. $+0.259 \pm 0.040 \text{ tC ha}^{-1} \text{ yr}^{-1}$ over 1985–2024 on the balanced set, whole profile, true observation years. Over the last interval alone it is $+0.117 \pm 0.066$ — a confidence interval that includes zero.

Annotation (p3): “Which rate? And which report?” Fixed here: the basis is now stated wherever a rate appears (balanced set $n = 310$, unweighted, whole profile, true years), because the same data give $+0.209$ on the pairwise set — a factor 1.8 on the headline. The reporting basis was decided on 2026-08-17 and every quoted rate in the paper must name it.

1.1.0.3 Why the levelling-off is the interesting part. A soil that is still accumulating and a soil that has stopped are the same stock and different futures. The second derivative, not the stock, is what an inventory projection turns on.

Do not present the observed rate as truth (annotation, p7). Three campaigns measured by three protocols over four decades give a rate with a real and partly unquantified comparability error: 1985 is a decade wide, its organic layer had to be reconstructed, and the short interval's CI includes zero. The paper states what was measured and how, and treats agreement between models and observations as consistency rather than as validation.

1.2 Why it matters for the national greenhouse gas balance

1.2.0.1 The soil term is the uncertain one. Finland's forest soil carbon flux enters the national inventory, and whether it is a growing sink or a saturating one changes the reported balance and the policy conclusions drawn from it. [□ LIT REVIEW: the GHG inventory chain: how the soil term is produced, reported and used; where our models sit in it](#)

1.2.0.2 What the inventory asks a model to do. Not to rank plots, but to reproduce a national trajectory over decades and project it forward. That is the quantity this paper evaluates, and saying so early prevents a later result (near-zero plot-level skill) from reading as failure.

1.3 The convention: soils initialised at steady state

1.3.0.1 It is near-universal, and rarely examined. The dominant practice across inventory applications is to start the soil at the equilibrium implied by contemporary litter input. [□ LIT REVIEW: steady-state initialisation across model lineages — Yasso, RothC, Century/DayCent, ICBM, CBM-CFS3; how each initialises, and how often the assumption is tested](#)

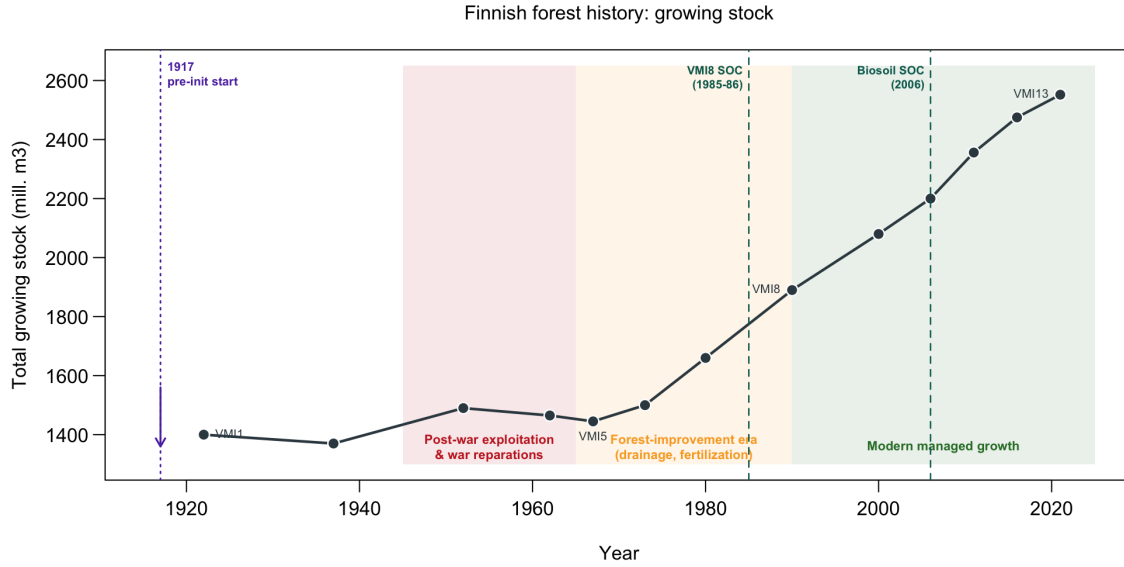
Annotation (p3, twice): “This will need literature review across multiple model lineages.” Acknowledged, not yet done — flagged here as a blocking dependency: the claim of near-universality cannot be made without it.

1.3.0.2 What steady state assumes. [□ LIT REVIEW: the theory paragraph \(annotation, p5\): what an equilibrium state is for a linear donor-controlled system, what it assumes about the constancy of inputs and climate, what properties follow — \$C = J/k\$, the loss of all timing information, and the fact that the assumption is about the *history*, not about the soil](#)

1.3.0.3 And what follows from it. An equilibrium start is already at the end of the accumulation. It cannot produce a rise, so it cannot show a rise levelling off; the very dynamic the inventory needs is the one the convention defines away.

1.4 Finnish forests are a managed landscape, not an equilibrium one

1.4.0.1 The history is the argument. Exploitation, then a century of managed regrowth: growing stock rose from roughly 1400 to 1775 million m³ between 1917 and 1985 and has risen further since. A soil integrating a century of that is not at equilibrium with today's litter. [► F10b_forest_history](#)



Source: Korhonen, Raty et al. (2024) Silva Fennica 58(4) art. 24045 (NF11-NF113)

Figure 2: F10b_forest_history

Annotation (p5): “This is load bearing and will also need proper literature revision.” Agreed — and it is the load-bearing claim of the whole paper. □ LIT REVIEW: Finnish forest management history and its soil-carbon consequences; the non-equilibrium initialisation problem (Peltoniemi 2004 as the direct antecedent); evidence on how long soils lag a change in input

1.5 Objectives

1.5.0.1 Three questions, in the order the paper weights them. (i) Does a transient, calibrated initialisation reproduce the observed national trajectory, and what does reproducing it require of the models? (ii) Does it matter, for extrapolation, where the calibration lands on the input/-turnover trade-off? (iii) What is left that none of the models can see, and what would it take to see it?

2 Materials and methods

2.1 Data

2.1.0.1 Plots and campaigns. Finnish NFI/BioSoil/Komeetta permanent plots; three SOC campaigns (1985–95, 2006, 2024); 456 calibration-ready plots, 1205 plot-year observations.

2.1.0.2 One homogenized SOC basis. Organic plus mineral to a per-plot variable depth cap plus a modelled deep tail, reproducing the official national stocks exactly. ► [F2_baseline](#)

2.1.0.3 Litter input. The Tupek product, tree litter including harvest residues and natural mortality, excluding understorey. ► [F1_inputs](#)

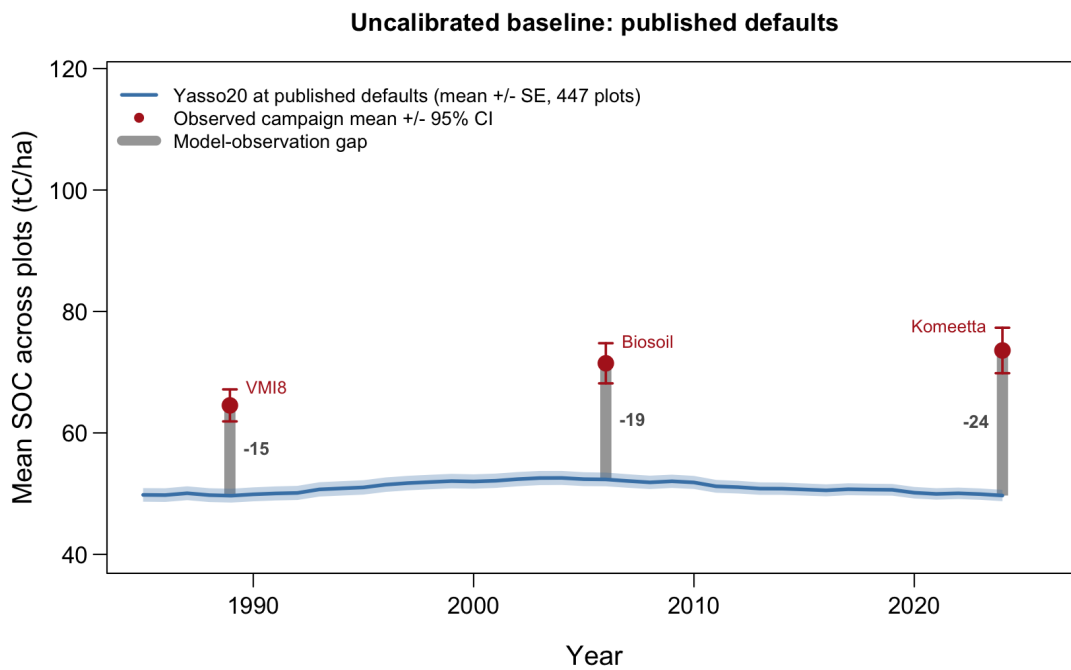


Figure 3: F2_baseline

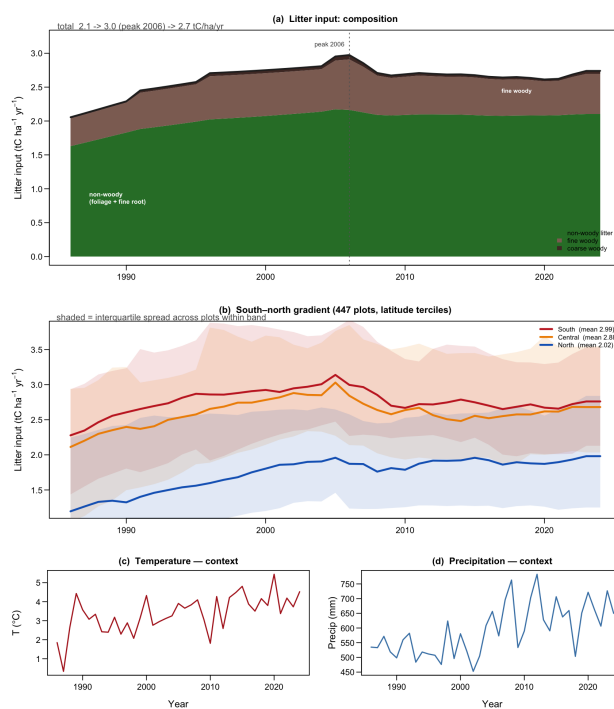


Figure 4: F1_inputs

2.2 The model ladder

2.2.0.1 Six models, one to five pools. SP1, TP2, TP3, Yasso07, Yasso15, Yasso20 — the simple models as an interpretable benchmark, the Yasso family as the operational standard.

Annotation (p9): “we are not really doing any structural uncertainty work ... We have the simpler structures as a benchmark, mainly, with Yasso as the proven standard.” **Adopted:** the ladder is described as a benchmark design, and the word “structural uncertainty” is used only where it is earned (the forward spread), never as a product claim.

2.3 Transient initialisation

2.3.0.1 The start becomes a parameter. Each draw spins up from 1917 under a reconstructed input ramp and is carried forward through all three campaigns; σ_{init} scales the 1917 input level. ► [F4_initialization](#)

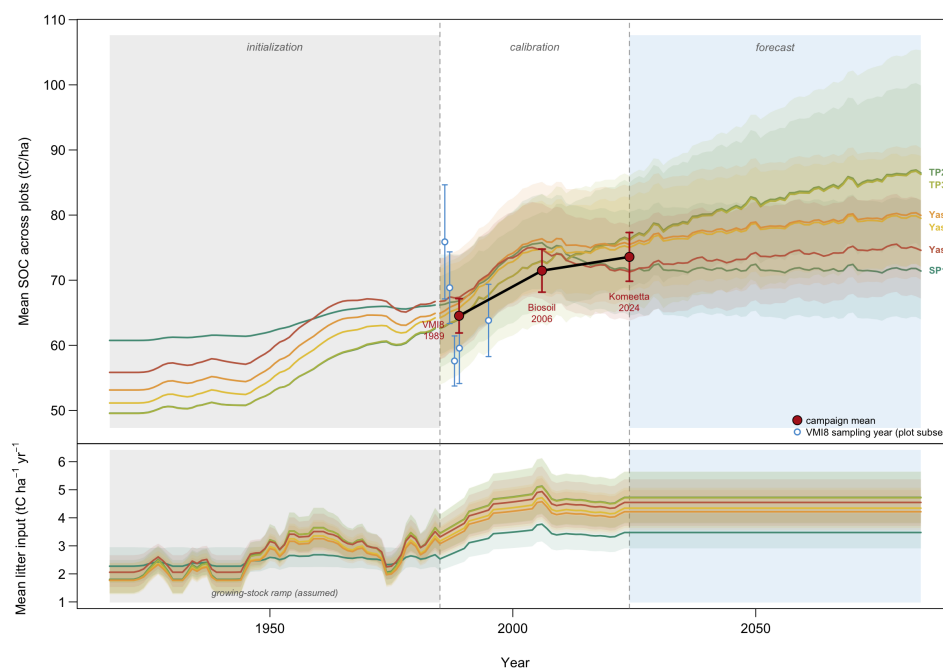


Figure 5: F4_initialization

2.4 A fair frame

2.4.0.1 Equal information, not equal parameter count. Yasso does not fit its decomposition rates, so the simple models do not either: intrinsic rates are externally anchored (ICBM), humification fractions are free, and every model calibrates the same auxiliary pair.

2.5 Calibration and error model

2.5.0.1 One likelihood, and it does not pretend plots are independent. Log-normal, with region, plot and campaign offsets marginalised at fixed variances.

2.6 Analyses

2.6.0.1 Three, matching the three objectives. Posterior predictive over the observed window; a forward experiment contrasting published and calibrated parameters under climate scenarios; and a residual analysis against site covariates.

3 Results and discussion

Combined, deliberately. Every result here needs its interpretation immediately: the level agreement means nothing without the ridge, and the ridge means nothing without what it costs. A split version is one toggle away (`\combinedfalse`) if a target journal requires it.

STRAND 1 — THE NATIONAL TRAJECTORY (*~55% of the paper*)

3.0.1 The models reach the observed levels

3.0.1.1 No model sits more than 3.5 tC ha^{-1} from any campaign mean. On the current calibration the ensemble reproduces all three campaign levels.

Annotation (p7): “I don’t get what this refers to.” The v2 text pointed at a “ 26 tC ha^{-1} too high at 1985” caveat that no reader could know about — it was an internal memory of an earlier run. **Deleted;** the sentence now states the present result only.

3.0.2 And they reproduce the rate, within the limits of the data

3.0.2.1 The ensemble brackets the observed rate, and two members land on it. Yasso07 $+0.258$ and Yasso15 $+0.250$ against an observed $+0.259$; TP2 and TP3 run stronger, SP1 and Yasso20 weaker. ▶ [F5_stock_change](#)

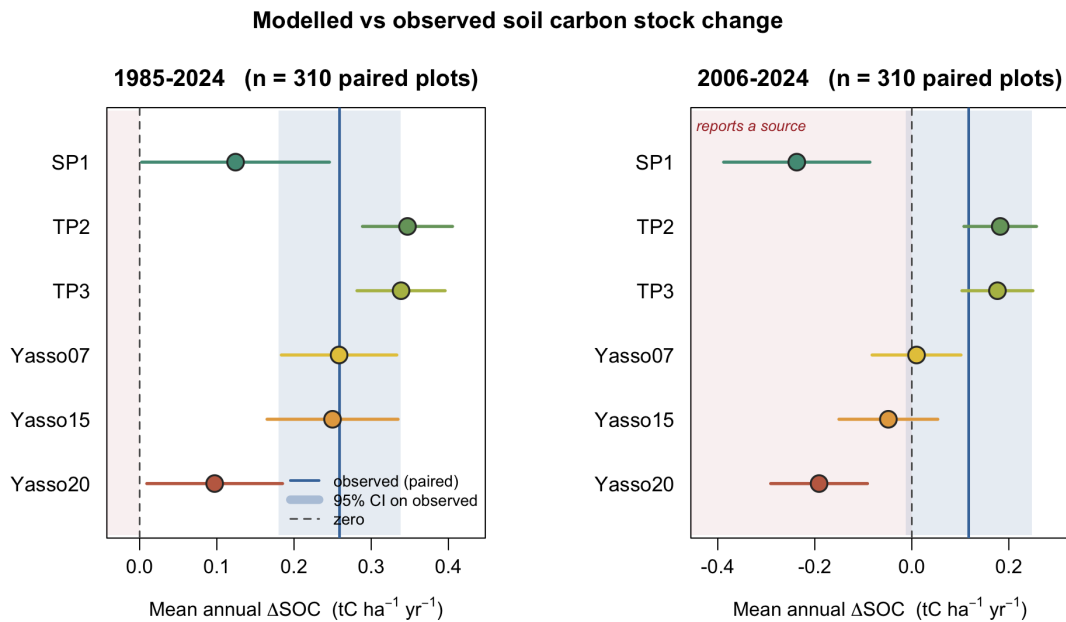


Figure 6: F5_stock_change

Stated as consistency, not as validation (annotation, p7). The observed rate carries its own comparability error, so “two models land on it” is reported as what was seen, with the campaign caveats attached, and no model is preferred on this basis.

3.0.3 The trajectory shape splits the ensemble three ways

3.0.3.1 Three behaviours, and they are legible. Yasso07 and Yasso15 rise and flatten; TP2 and TP3 rise and keep rising; SP1 and Yasso20 peak in the late 2000s and decline. The same split is visible in the per-plot deliverable. ★ [TO BUILD: F3b — trajectory shape, three groups, explicitly grouped](#)

Annotation (p7): “Try it, but then we have to look at it together before including; current Fig 6 in this text could be enough already.” Kept as [TO BUILD](#) — and note that the grouping is already visible without a new figure in the NextGenC trajectory panel, which may make F3b unnecessary. Decide when it exists.

3.0.4 This is what an equilibrium start forecloses

3.0.4.1 The shape is the point. Accumulation bending over is a statement about timing, and timing is precisely what an equilibrium initialisation discards. Reproducing it is the paper’s first result.

3.0.5 Structure is nearly invisible in fit

3.0.5.1 Skill does not climb the pool ladder. Calibration R^2 0.016–0.021; holdout $R^2 \leq 0.0012$; holdout RMSE spans 3.6%, and the best and worst belong to a five-pool and a three-pool model. ► [T1_model_summary](#) ► [F6b_skill_vs_complexity](#)

Model	Pools	Free params	R^2 (cal)	R^2 (hold)	RMSE (hold)	Eff. flux $\sigma_{\text{inp}} \bar{J}$	σ_{init}
SP1	1	6	0.017	0.000	47	3.2	0.84
TP2	2	7	0.016	0.000	48	4.3	0.50
TP3	3	9	0.015	0.000	49	4.3	0.49
Yasso07	5	20	0.019	0.000	48	4.0	0.52
Yasso15	5	26	0.021	0.001	47	3.9	0.54
Yasso20	5	26	0.016	0.000	47	4.2	0.59

3.0.5.2 So fit cannot arbitrate between these structures. Whatever separates them, it is not the calibration data.

3.0.6 The data pin a level, and little else

3.0.6.1 Convergence is not identifiability. R-hat ≤ 1.008 everywhere, yet the transfer fractions are prior-pinned, ridge-correlated and carry under a nat. ► [F8b_fraction_ridges](#) ► [F6_obs_vs_pred_holdout](#)

3.0.7 What fitting the trajectory requires: more input, and faster turnover

3.0.7.1 The calibration lands away from the published parameterisations. Effective litter flux 3.19–4.34 tC ha^{−1} yr^{−1}, and intrinsic MRT 24.6 / 27.1 / 22.6 yr for Yasso07/15/20 against published 33.5 / 30.5 / 22.1. ► [F12_mrt_yasso](#)

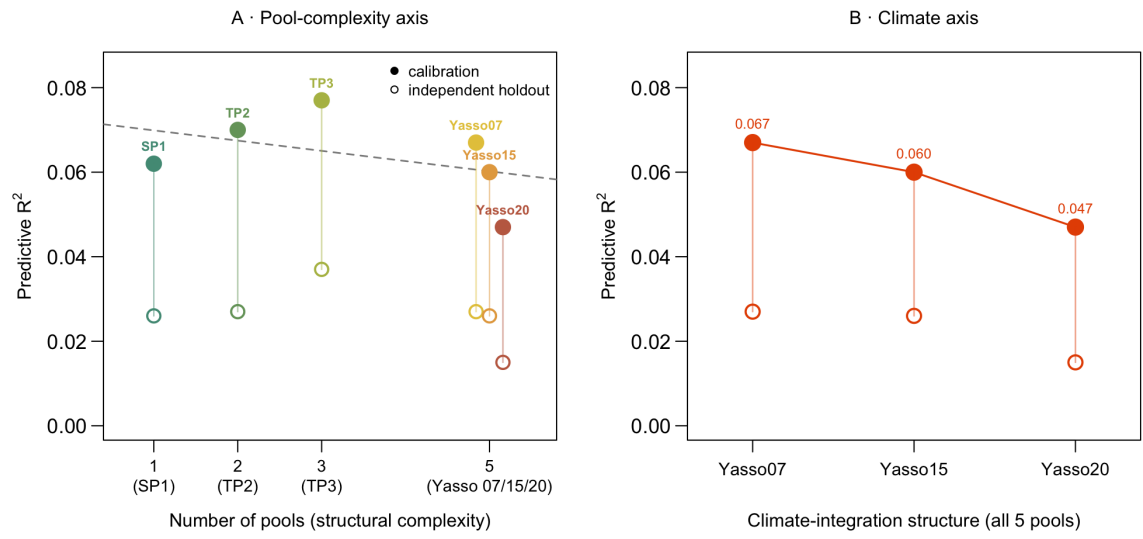


Figure 7: F6b_skill_vs_complexity

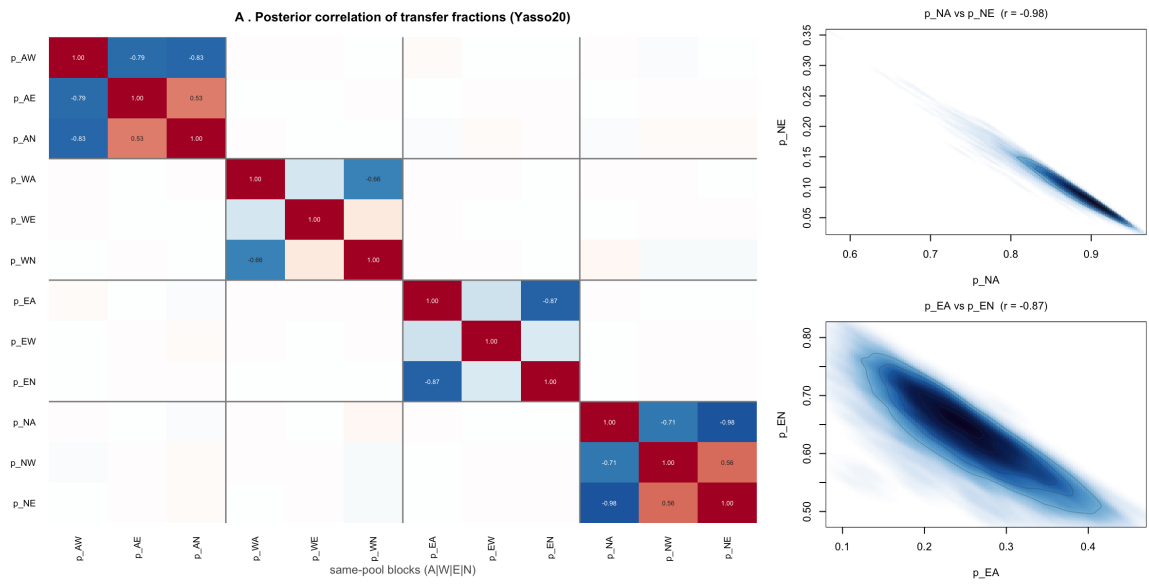


Figure 8: F8b_fraction_ridges

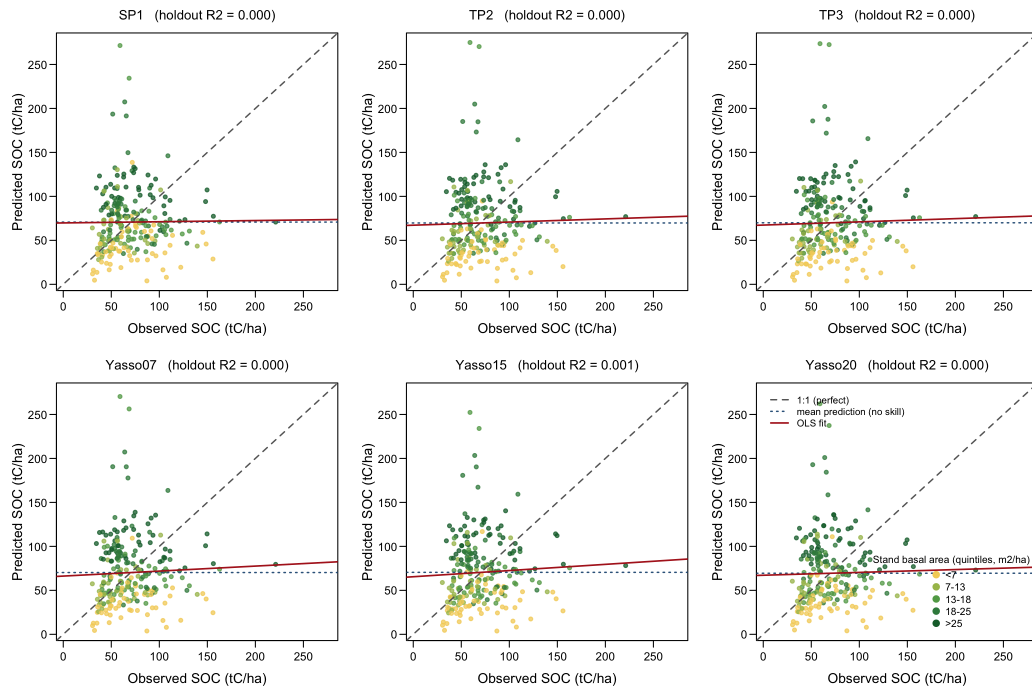


Figure 9: F6_obs_vs_pred_holdout

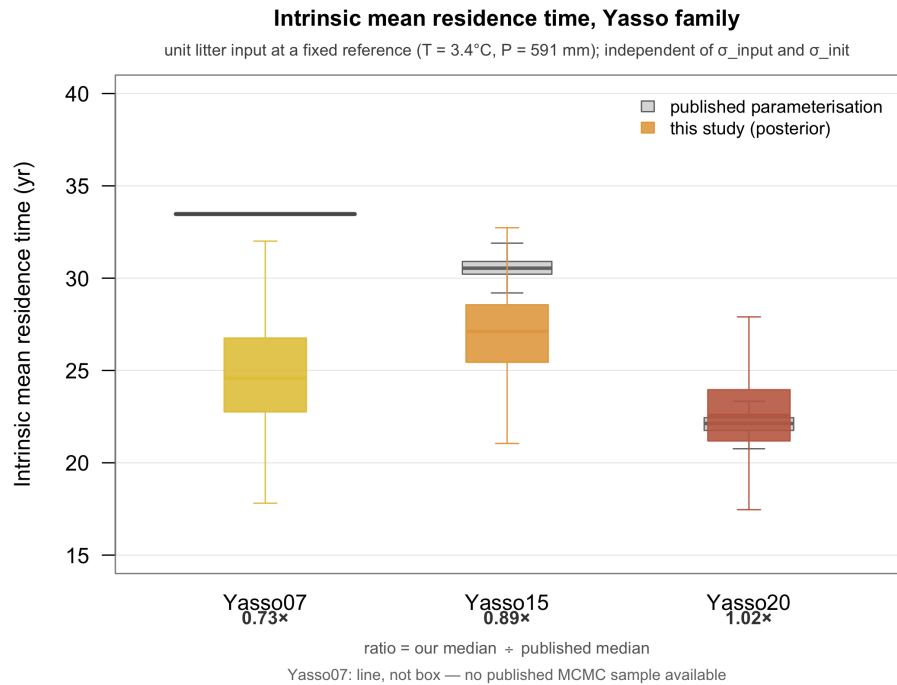


Figure 10: F12_mrt_yasso

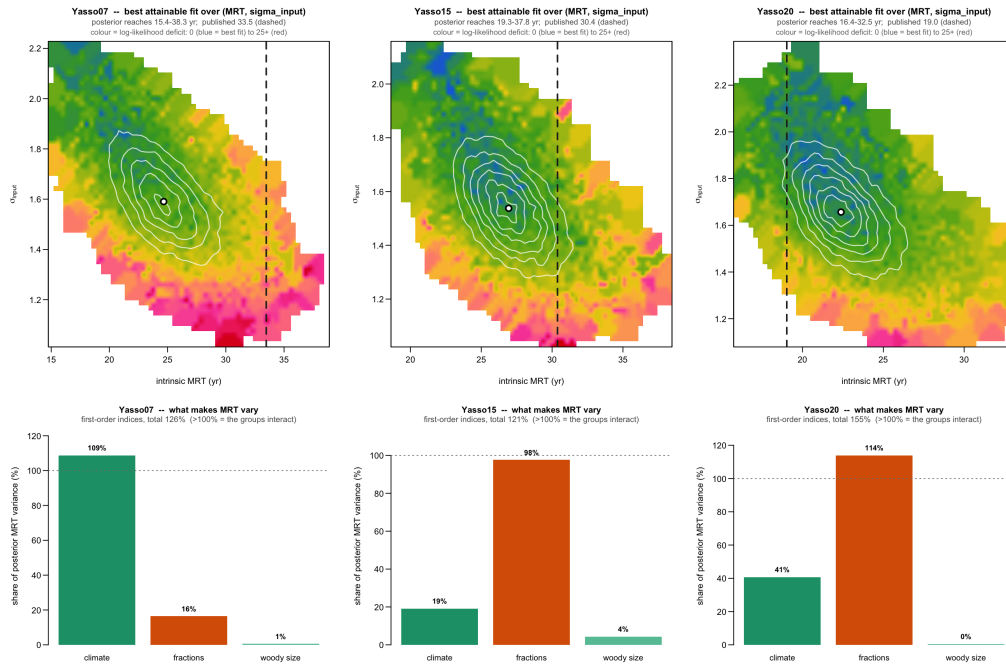


Figure 11: F14_mrt_ridge

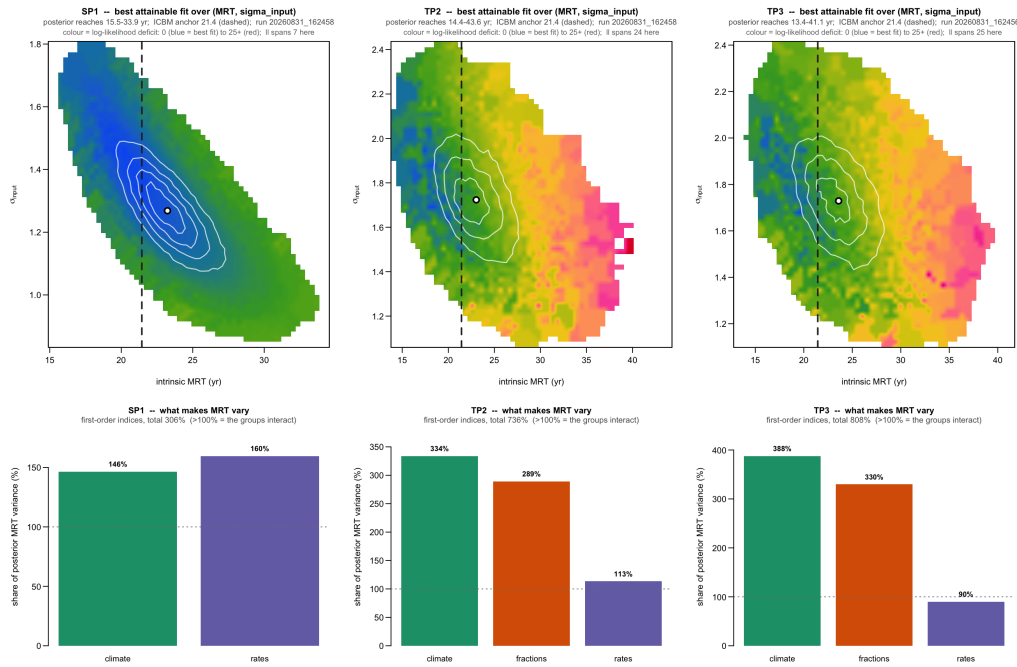


Figure 12: S13_mrt_ridge_benchmark

3.0.7.2 These are one displacement, not two. The level constrains $\text{MRT} \times \sigma_{\text{input}}$, so faster-with-more and slower-with-less fit alike; the posterior rides that ridge. ► [F14_mrt_ridge](#) ► [S13_mrt_ridge_benchmark](#)

Annotation (p12), on “a steady-state start hides exactly this”: “*Steady state still has the same problem of equifinality ... with the trajectories the ridge on the likelihood surface should narrow a bit ... no need to overdo this surprise thing.*” **Rewritten.** The claim is now the weaker and defensible one: *the ridge exists under either initialisation; a transient start narrows it, because the approach rate carries information the equilibrium condition discards.* Measured, the narrowing is partial — MRT and σ_{input} still correlate -0.64 to -0.73 . The “surprise” framing is gone; the displacement is reported as a result, once.

Annotation (p12), on Liski: “I don’t think so, we are correcting even higher.” Confirmed and corrected. Liski 2006 implies $\sigma_{\text{input}} \approx 1.27$; our effective correction runs 1.27 – 1.73 across the six. Liski is the **floor** of our range, not a match — so the sentence becomes “an independent estimate from the same country, period and model family sits at the bottom of the range we infer”, which is a weaker corroboration and an honest one.

3.0.8 What might be missing from the input flux

Annotation (p12), the longest one: “*The input functions have a lot of trust in the field ... they do work (I think because the previous fitting compensates) ... there needs to be a very extensive review of the supporting evidence, with hypotheses well grounded in literature about what we might be missing in the fluxes (first things to examine are fine root MRT, exudates and mycorrhizae) ... also missing pioneer vegetation.*” **This section is the answer to that note, and it is the main place where the paper stops being about models.**

3.0.8.1 The candidate is not that the litter model is wrong, but that it is incomplete. Litter input products are trusted because they have worked for decades — and one reason they work is that a calibration free to adjust rates can absorb a missing flux without visible misfit. A transient calibration on a fixed rate anchor removes that compensation, and what is missing then has to appear somewhere.

3.0.8.2 Belowground turnover. Fine root lifespan is the largest single lever on belowground litter and is poorly constrained. □ [LIT REVIEW: fine root MRT in boreal conifers: isotopic vs minirhizotron estimates, and the factor-of-several disagreement between methods](#)

3.0.8.3 Rhizodeposition. Exudates bypass the litter pathway entirely and are absent from inventory litter models by construction. □ [LIT REVIEW: exudate fluxes as a fraction of NPP in boreal forest](#)

3.0.8.4 Mycorrhizal necromass. Increasingly implicated as a dominant precursor of stable soil carbon in boreal systems, and entirely outside the litter product. □ [LIT REVIEW: mycorrhizal contribution to boreal SOC formation — the Clemmensen line of evidence and what has followed it](#)

3.0.8.5 Understorey. Excluded from the product by construction, and its share rises strongly toward the north — so a single national multiplier must be wrong in opposite directions at the two ends of the country. ► [SX_input_vs_literature](#)

3.0.8.6 Early-succession vegetation. Pioneer species after disturbance contribute litter that inventory-based models, keyed to standing merchantable biomass, do not represent. □ [LIT REVIEW: pioneer/early-succession litter contributions after clear-cut and after fire](#)

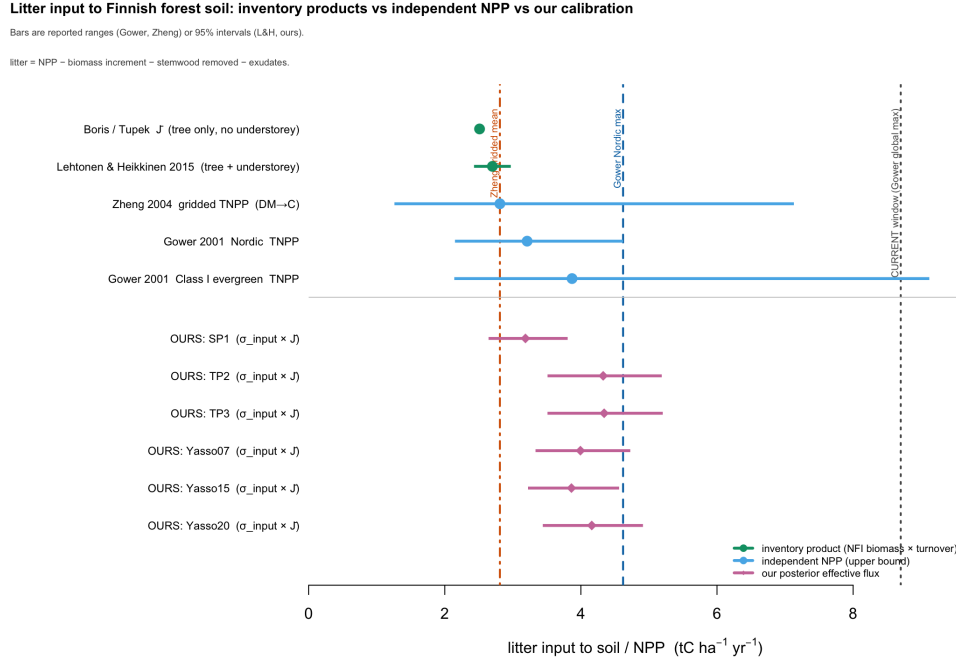


Figure 13: SX_input_vs_literature

This section must not become an argument that we are right. Each mechanism is a hypothesis for where an apparently missing flux could come from, and several are not separable with our data. The honest form is: the calibration says the input is larger than the product; here are physically plausible sources; testing them is not this paper's job.

STRAND 2 — DOES RIDGE POSITION MATTER? ($\sim 25\%$)

Annotation (p17): “For what? Do you mean if it has consequences for extrapolation, for example?” Yes — and the section is now titled that way. v2’s “does the ridge position matter?” left the reader to guess the predicate. It matters *for extrapolation*: two calibrations that fit the observed window identically can project differently.

3.0.9 The question the ridge forces

3.0.9.1 The data cannot choose a point on the ridge, so the choice must be judged by its consequences. If moving along it changes nothing we can observe, the displacement is bookkeeping. If it changes the projection, it is a result with consequences for anyone using these models.

3.0.10 The test: published parameters against ours, forward only

3.0.10.1 No re-calibration, one input parameter refitted. Both arms are pushed through the same machinery under ramped scenarios; both match the observed window at least as well as each other, so any divergence afterwards is extrapolation. ► [F15_forward_arms](#)

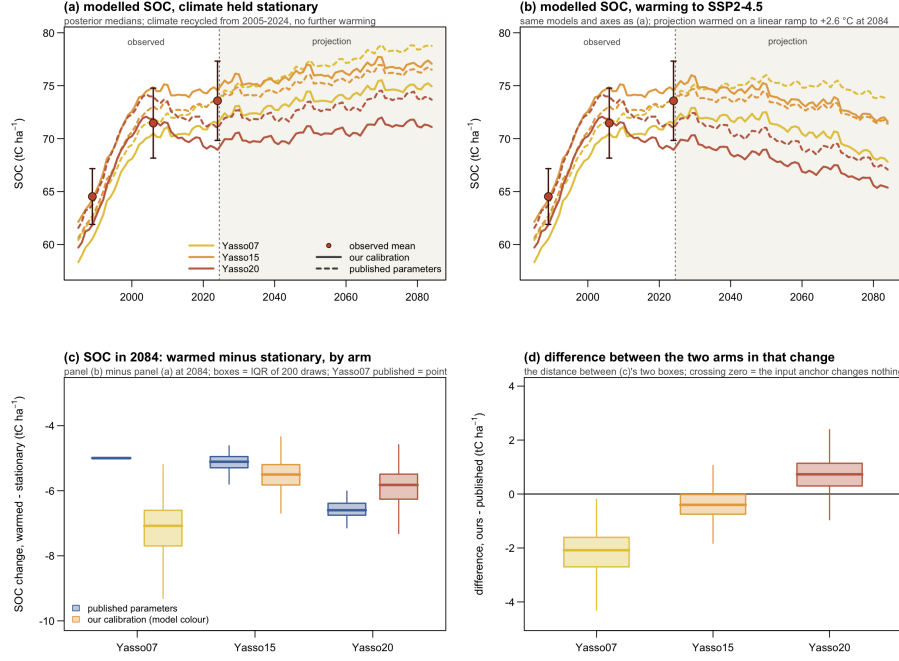


Figure 14: F15_forward_arms

3.0.11 It matters for one structure and not the others

3.0.11.1 Only Yasso07 separates. $-2.08 [-3.57, -0.79]$ tC ha⁻¹ at SSP2-4.5; Yasso15 and Yasso20 do not separate at all.

3.0.11.2 And the mechanism is structural. Yasso07 applies a single climate modifier to every pool, so turnover and temperature sensitivity are tied to the same parameter and moving along the ridge moves both. Yasso15 and Yasso20 carry pool-specific modifiers, and the pools that *hold* the carbon respond almost identically in both arms. ► [A1_forward_appendix](#)

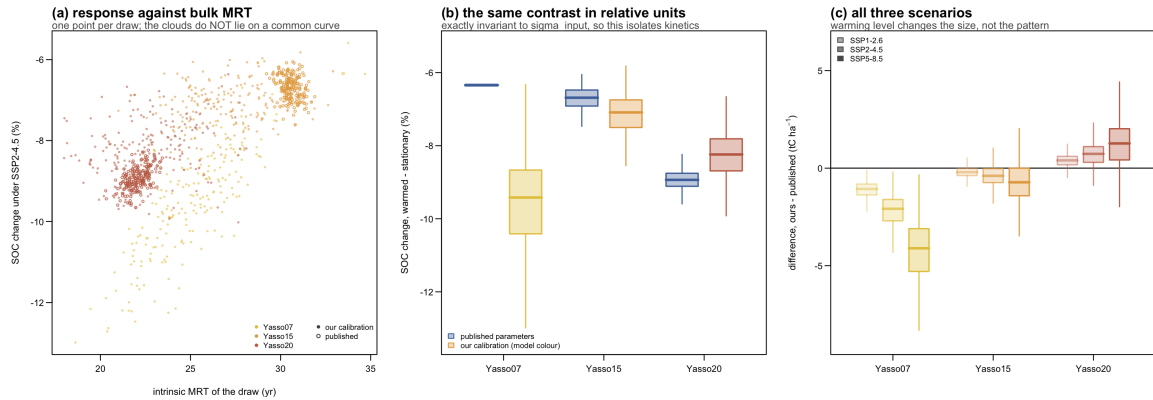


Figure 15: A1_forward_appendix

3.0.11.3 The framing that survives. Yasso15 and Yasso20 are not shown to be right; they are shown to be **resilient** to an input assumption the data cannot settle. That is a usable statement for

an inventory, and a narrower one than “our calibration changes the forecast”.

The main forecast contains no climate change, and the paper must say so. The projection recycles the last twenty observed years and freezes litter at 2024, so its sink is relaxation toward equilibrium, not a response. The scenarios appear only in this side arc, as an instrument for testing ridge position. Doing it properly would need productivity to respond to warming as well.

STRAND 3 — WHAT NONE OF THE MODELS CAN SEE ($\sim 20\%$, and the last part brief)

Annotations (p16, two): “*This seems about local variance, point variance, rather than the overall trend ... This is a separated topic. I would put it lower in priority ... we are just offering this analysis to raise new research questions ... Our main story is about the global, national variance.*” **Restructured accordingly:** this is no longer a subsection inside the input argument (where v2 put it, and where it did read as evidence for the input claim). It is its own strand, last, at $\sim 20\%$, and it is framed as a question raised rather than a gap filled.

3.0.12 The models order the national trajectory but not the plots

3.0.12.1 Two statements that read as a contradiction only if separated. The ensemble reproduces the national mean trajectory and explains almost none of the between-plot variance. Both are true, and the second is a statement about spatial resolution, not about the trajectory.

3.0.13 What is left behind is recoverable, not irreducible

3.0.13.1 Roughly half of the residual variance is predictable from covariates already in hand. A random forest on site variables explains 44–65% of what the models leave (out-of-bag), with observation reliability around 0.49 — so this is not a noise ceiling. ▶ [F11_rf_heatmap](#)

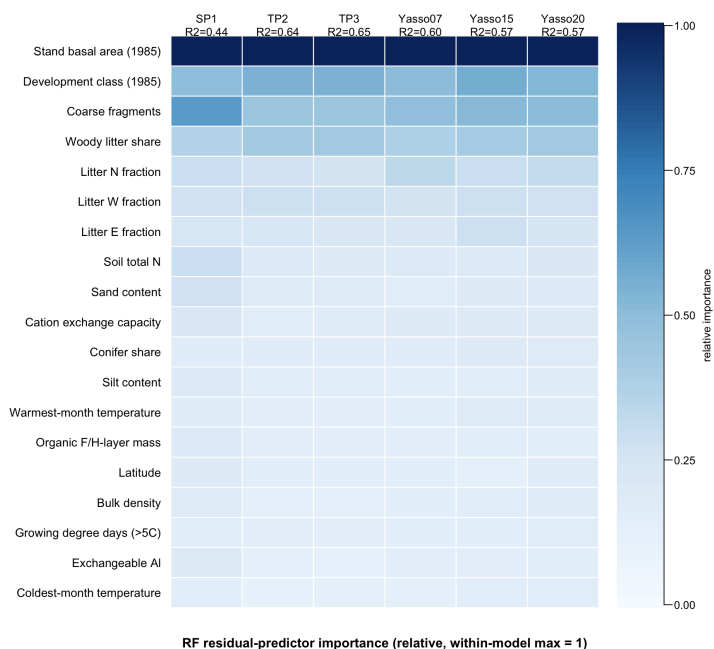


Figure 16: F11_rf_heatmap

3.0.13.2 The same covariates lead in all six models. Stand basal area first, then development class and coarse-fragment content; the importance is diffuse rather than concentrated in one term.

3.0.14 What that suggests, briefly

3.0.14.1 Two of the leaders are productivity proxies and one is soil-physical. A productivity proxy leading everywhere is what a spatially varying input level would look like; a soil-physical variable leading is something no model in the ensemble has a term for. **We name the observation and stop there** — the method that would exploit it is not this paper’s to choose.

4 Conclusions

Annotation (p22): “The policy deliverable should not be a real conclusion, it is more a nice bonus product. The conclusions ... are more about the opening stories, their resolutions.”
Rebuilt exactly that way: the conclusions below answer the three opening questions in the order of the weight budget, and the deliverable is demoted to a closing paragraph.

4.0.0.1 A transient, calibrated start reproduces a dynamic the conventional one cannot. The observed accumulation and its levelling-off are recovered by all six models at the level and by the ensemble in rate, which an equilibrium initialisation forecloses by construction.

4.0.0.2 Reproducing it requires more litter input, and faster turnover, than the published parameterisations carry. These are one displacement along a trade-off the data cannot resolve, and the most plausible reading is not that the models are wrong but that the input flux is incomplete — with belowground and non-tree sources the candidates.

4.0.0.3 Where the calibration lands on that trade-off reaches the projection, but only for some structures. A single climate modifier ties turnover to temperature sensitivity; pool-specific modifiers do not. Models of the second kind are more robust to an assumption that cannot currently be tested.

4.0.0.4 And a large, recoverable part of the plot-scale variance is invisible to every model here. It is a well-posed question for the next study rather than a limitation of this one.

4.0.0.5 Bonus product, not conclusion: the maps. The calibrated ensemble yields gridded SOC and its uncertainty for Finland, which is useful to the inventory and is offered as a deliverable rather than as the paper’s finding. ► [ENSEMBLE_SOC_mean](#)

Open decisions and what is still missing

4.0.0.6 Structural. Whether Results and Discussion stay combined (currently yes). Whether F3b is built or the NextGenC panel suffices. Whether the arc diagram’s thread vocabulary is retired.

4.0.0.7 Blocking literature reviews. Steady-state initialisation across model lineages; the theory of equilibrium initialisation; Finnish management history; and the whole of §3.0.8 — fine root MRT, exudates, mycorrhizal necromass, pioneer vegetation.

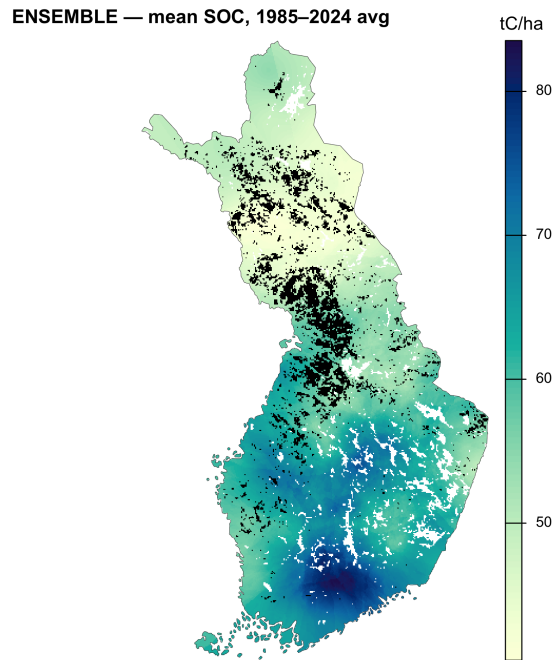


Figure 17: ENSEMBLE_SOC_mean (NextGenC deliverable)

4.0.0.8 Still not started. Citations and a .bib; title and abstract; the coauthor block; figure placement in the manuscript.

4.0.0.9 Unresolved from earlier sessions. The 1917 stock floor is still an unsourced placeholder, and the relaunch of the six calibrations after the Fortran guard change has not been done.